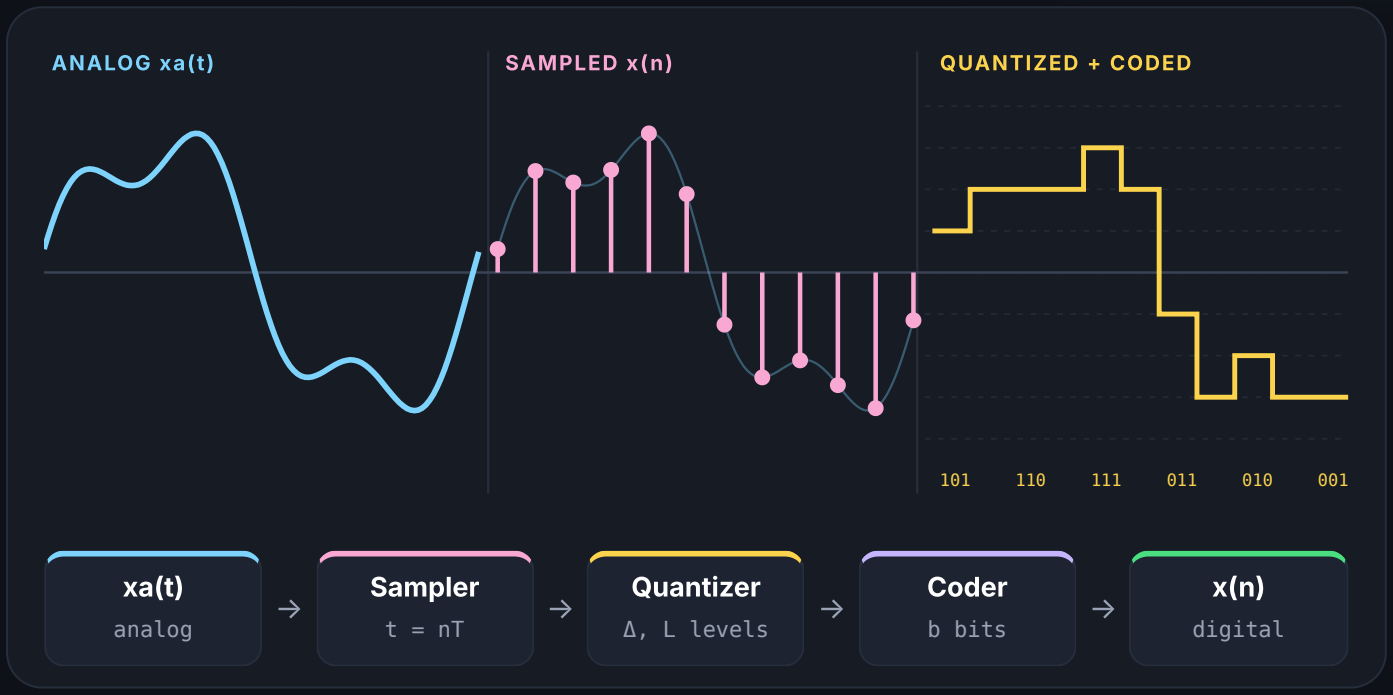
CHAPTERS

01·02 Repeated Theory in the Exams

Repeated Theory · Ch 1 & 2

Every theory question that came in the previous midterms, finals and quizzes — the doctor's model answers, most repeated first



13 Chapter 1 questions

3 Chapter 2 questions

10 repeated 2+ times

24 previous exams & quizzes

CHAPTER 1
Sampling & quantization
Lectures 2 · 3 · 4

CHAPTER 2
DFT & FFT
Lectures 5 · 6

ORDER
Most repeated first
exam count on every card

Lecturer: Dr. Samir M. Kamal
Modern Academy for Engineering & Technology

DSP / ch01 / contents

Contents

Repeated theory · Ch 1 & Ch 2 · 16 questions

1 Chapter 1 Introduction to DSP

01 **Chapter 1**
Repeated in exams (2 times or more), most repeated first — 8 questions — p. 3

02 **Chapter 1**
Came once in the exams — 5 questions — p. 13

2 Chapter 2 The DFT and the FFT

03 **Chapter 2**
Repeated in exams (2 times or more), most repeated first — 2 questions — p. 19

04 **Chapter 2**
Came once in the exams — 1 questions — p. 21

How to study from this file

Chapter 1 then Chapter 2. In each chapter the questions that came 2 times or more come first, ordered by the number of exams, then the questions that came once.

SECTION 01

Chapter 1

Repeated in exams (2 times or more), most repeated first

Questions — most frequent first

- ☐ Q9 In detail, what are the main characteristics of the discrete-time sinusoidal signals... p. 3
- ☐ Q15 Find the relationship between the frequency of the analog signal f, the frequency of... p. 5
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- ☐ Q16 In sampling process, derive the sampling condition Fs ≥ 2f max (avoiding aliasing). p. 7
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- ☐ Q22 Check (✓) for the right statements and (×) for the wrong statements and correct the ... p. 10
- ☐ Q1 Define the following items: Signal, System, and Signal Processing (give an example f... p. 11

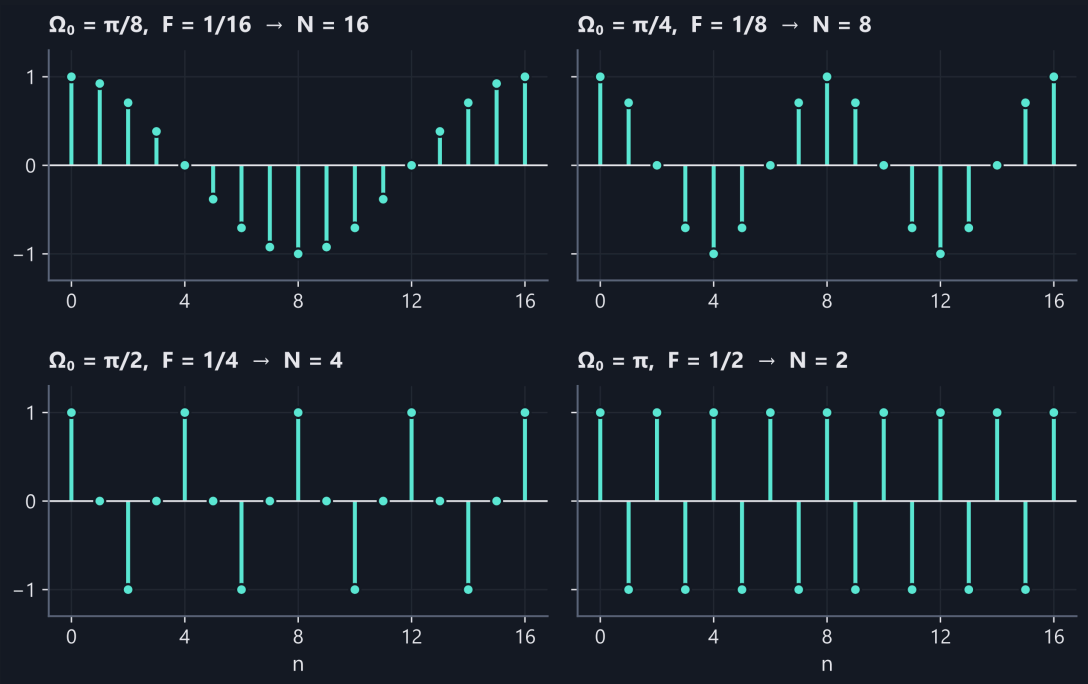
Q9 ● Frequent 9×

In detail, what are the main characteristics of the discrete-time sinusoidal signals?

$x(n) = A\cos(\Omega n + \theta) = A\cos(2\pi F n + \theta)$, where n is the sample number, Ω is in radians/sample and F in cycles/sample.

1. **(a)** A discrete-time sinusoid is **periodic only if its frequency F is a rational number** ($F_o = k/N$).
2. **(b)** Discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are **identical**. Frequencies in $-\pi \le \Omega \le \pi$ ($-1/2 \le F \le 1/2$) are unique; all frequencies $|\Omega| > \pi$ are **aliases**.

3. **(c)** The **highest rate of oscillation** is attained when Ω = π (or Ω = −π), i.e. F = 1/2 (or F = −1/2).



Key idea الـ discrete sinusoid تبقى periodic بس لو F rational. الترددات اللي بينها 2π بتدّي نفس الإشارة، وأسرع تذبذب بيكون عند ω = π.

Source — from your PDFs

Lecture **LEC 3** · Sec. 1.5.2 · p. 9

Came in Final 11-6-2023 · Final 7-6-2023 · Q4-a · Final 12-1-2019 · Final 2-1-2017 · Final 23-12-2017 · Midterm 26-7-2026 · Midterm 31-7-2023 · Midterm OLD · Final theory notes

Q15 · Frequent · 8×

Find the relationship between the frequency of the analog signal f, the frequency of the discrete-time signal F, and the sampling frequency Fs.

Consider xa(t) = A cos(2πft + θ). Sampling at Fs = 1/T (t = nT = n/Fs):

$$x(n) = x_a(nT) = A\cos(2\pi fnT + \theta) = A\cos(2\pi nf/F_s + \theta) \quad (1.32)$$

Comparing with x(n) = A cos(2πFn + θ) (1.14):

$$F = f / F_s \quad (1.33) \qquad \text{or} \qquad \Omega = \omega T \quad (1.34)$$

Key idea لما نعوّض t = nT بنلاقي F = f/Fs، يعني التردد الرقمي هو التردد الأنالوج مقسوم على الـ sampling rate.

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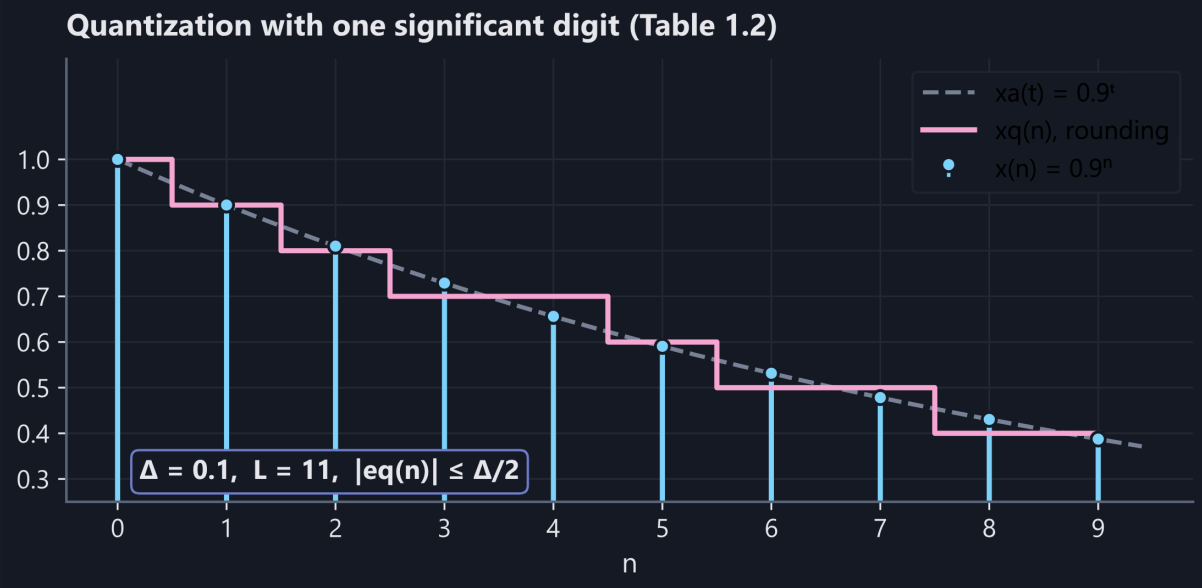
Lecture LEC 4 · Sec. 1.6.1 · p. 8

Came in Final 18-1-2024 · Q1-a · Final 12-1-2019 · Q2-a · Final 2-1-2017 · Q2-a · Final 23-12-2017 · Q2-a · Final 28-12-2019 · Q1-a · Midterm 12-11-2025 · Q1-a · Midterm 2 · Q2-a · Final 8-9-2021

Q18 ● Frequent 8×

In A/D, explain in detail the quantization process of continuous-amplitude signals.

- **Quantization:** the process of converting a discrete-time continuous-amplitude signal into a digital signal by expressing each sample value as a finite number of digits.
- **Quantization error:** eq(n) = xq(n) − x(n) (1.53), where xq(n) = Q[x(n)].
- The values allowed are the **quantization levels**; the distance between two successive levels is the **step size Δ**.
- In rounding: $-\Delta/2 \le e_q(n) \le \Delta/2$ (1.54), and $\Delta = (x_{max} - x_{min})/(L - 1)$ (1.55).
- Increasing the number of levels L decreases Δ, so the quantization error decreases and the accuracy increases.



Key idea كل sample بيتقرّب لواحد من L levels، والمسافة بينهم Δ = (xmax − xmin)/(L − 1)، وأقصى error مع الـ rounding هو Δ/2.

Key words Quantization Quantization error

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.6.3 · p. 23

Came in Final 7-6-2023 · Q1-a | Final 17-8-2026 · Q1-a | Final 4-3-2021 · Q1-a | Final 8-9-2021 · Q1-a | Midterm 2 · Q1-b | Quiz 1 · Section | Midterm OLD | Final theory notes

Q16 ● Frequent 6×

In sampling process, derive the sampling condition $Fs \geq 2f_{max}$ (avoiding aliasing).

Since the discrete-time frequency must satisfy $-1/2 \leq F \leq 1/2$ and $F = f/Fs$:

$$-1/2 \leq f/Fs \leq 1/2 \rightarrow -Fs/2 \leq f \leq Fs/2 \quad (1.37)$$

$$f_{max} = Fs/2 \rightarrow \mathbf{Fs \geq 2f_{max}}$$

Thus to avoid the problem of aliasing, Fs is selected so that $Fs > 2f_{max}$ (1.47).

Key idea عشان كل component تتمثل صح لازم $|F| \leq 1/2$، يعني $f \leq Fs/2$، يعني لو $Fs \geq 2fmax$. ده ما حصلش، الترددات العالية بتعمل aliasing.

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.6.1 · p. 8

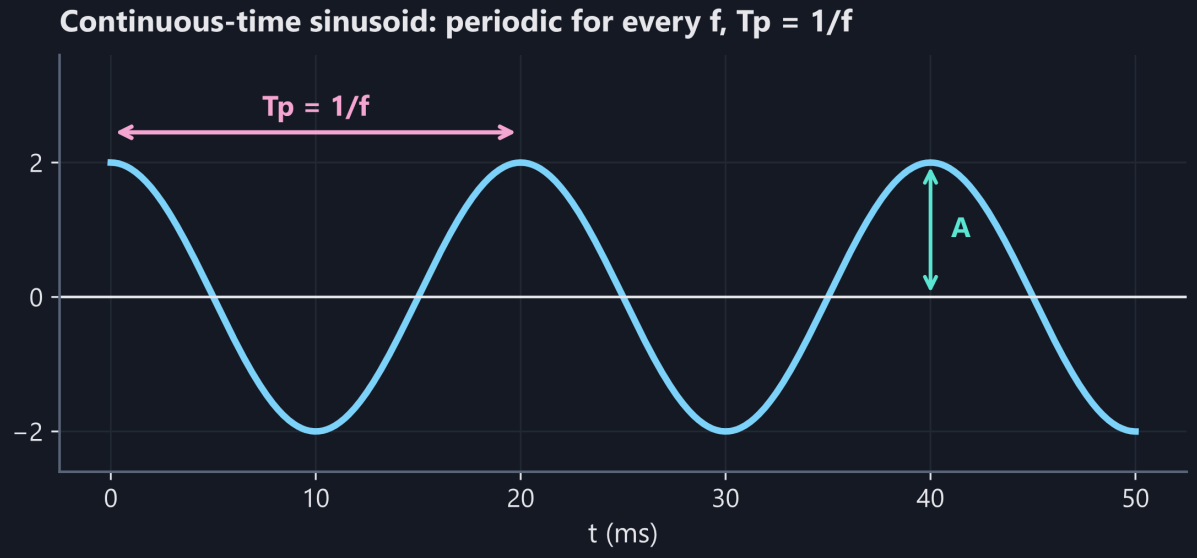
Came in Final 20-1-2025 · Q4-a | Final 27-6-2021 · Q1-a | Midterm 26-7-2026 · Q2-a | Midterm 31-7-2023 · Q2-a | Midterm 2 · Q1-a | Quiz 1 Summer 26 · Q1-a

Q8 ● Frequent 4×

In detail, what are the main characteristics of the continuous-time sinusoidal signals?

$x_a(t) = A\cos(\omega t + \theta) = A\cos(2\pi f t + \theta)$, where A is the amplitude, $\omega = 2\pi f$ the frequency in rad/s, f in Hz, and θ the phase in radians.

1. **(a)** For every fixed value of the frequency f, $x_a(t)$ is **periodic**: $x_a(t + T_p) = x_a(t)$, where $T_p = 1/f$ is the fundamental period.
2. **(b)** Continuous-time sinusoidal signals with distinct (different) frequencies are themselves **distinct**.
3. **(c)** Increasing the frequency f results in an **increase in the rate of oscillation** of the signal; f can be increased without limit.



Key idea الـ continuous sinusoid دايمًا periodic، وكل تردد مختلف بيدّي إشارة مختلفة، وكل ما التردد يزيد الإشارة بتتذبذب أسرع.

Key words periodic

Source — from your PDFs

Lecture **LEC 3** · Sec. 1.5.1 · p. 3

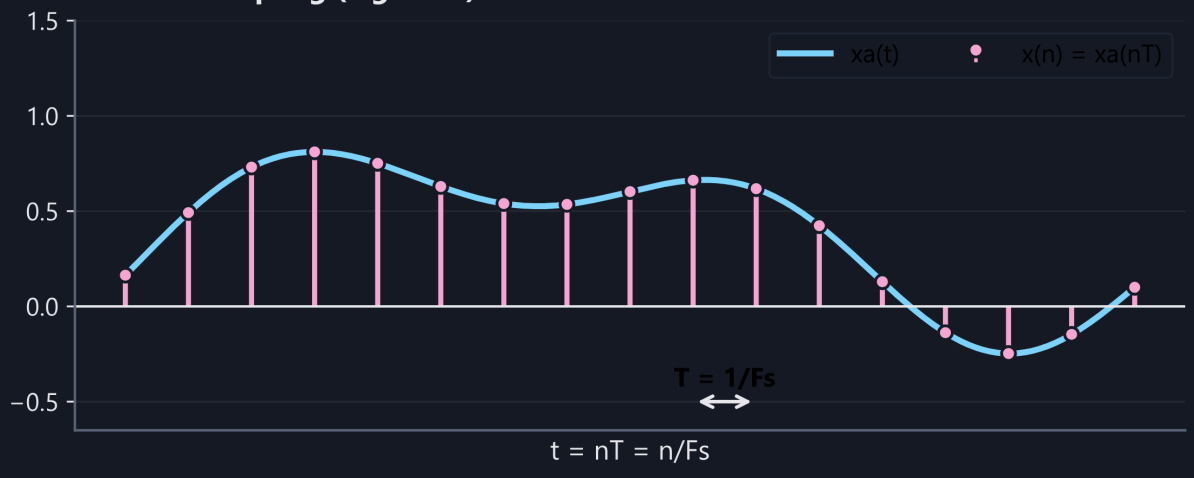
Came in Final 20-1-2025 · Q3-a | Final 9-6-2026 · Q1-a | Quiz 1 Summer 26 · Q1-b | Midterm OLD

Q14 ● Frequent 4×

In A/D, explain in detail the sampling process of analog signals.

- We use periodic (uniform) sampling, described by the relation **x(n) = xa(nT)**, $-\infty < n < \infty$ (1.29), where x(n) is obtained by taking samples of xa(t) every T seconds (Fig. 1.11).
- The time interval T between successive samples is the **sampling period**; its reciprocal **Fs = 1/T** is the sampling rate (samples/s) or sampling frequency (Hz).
- Periodic sampling relates the time variables t and n: **t = nT = n/Fs** (1.30). As a consequence, the frequencies are related by F = f/Fs (1.33).
- Sampling does not result in a loss of information if the bandwidth is finite and Fs > $2f_{max}$.

Periodic sampling (Fig. 1.11)



Key idea بناخد sample كل T ثانية: x(n) = xa(nT)، و T = 1/Fs.

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.6.1 · p. 8

Came in Midterm 28-11-2021 · Q1-a | Midterm 8-12-2020 · Q1-a | Midterm 1 Model-2 · Q1-a | Midterm OLD

Q22 ● Frequent 4×

Check (✓) for the right statements and (×) for the wrong statements and correct the mistakes.

1. In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases. **(×)**
   Correct: when Δ **decreases**, the accuracy increases.
2. Nyquist frequency (F_N) is the minimum sampling frequency. **(✓)**
3. To reconstruct x(t) from x(n), the sampling rate Fs must be less than the frequency of x(t), f_o. **(×)**
   Correct: Fs must be **greater than $2f_o$**.

Key idea Δ أصغر معناها دقة أعلى، فالجملة الأولى غلط. الـ Nyquist rate هو أقل sampling rate، فدي صح. وعشان نرجّع الإشارة لازم Fs > 2f0، فالأخيرة غلط.

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.6 · p. 3

Came in Final 12-1-2019 · Q5 | Final 23-12-2017 · Q5-b | Final 27-6-2021 · Q4 | Quiz 1 Summer 26 · Q3

Q1 3×

Define the following items: Signal, System, and Signal Processing (give an example for each).

- **Signal:** is defined as any physical quantity that varies with time, space, or any other independent variable(s). Mathematically, we describe a signal as a function of one or more independent variables.
  Examples: s1(t) = 5t (varies linearly with t), s2(t) = $20t^2$ (varies quadratically with t), and s(x, y) = 3x + 2xy + $10y^2$ (two independent variables, e.g. the spatial coordinates in a plane). A speech signal cannot be described by such an expression; a segment of it is represented as a sum of several sinusoids of different amplitudes, frequencies and phases.
- **System:** is defined as a physical device that performs an operation on a signal.
  Example: a filter used to reduce the noise and interference corrupting a desired information. If the operation is linear, the system is called linear; if it is nonlinear, the system is nonlinear.
- **Signal processing:** when we pass a signal through a system, as in filtering, we say that we have processed the signal.
  Example: filtering the noise and interference from the desired signal. Digital processing may be performed by digital hardware (logic circuits), by software, or by a combination of both.

Key idea الـ Signal أي كمية فيزيائية بتتغير مع متغير مستقل زي الوقت. الـ System هو الجهاز اللي بيشتغل على الـ signal. والـ Signal Processing إننا نعدّي الـ signal على system، زي الـ filter.

Key words: Signal | System | Signal processing

Source — from your PDFs

Lecture	**LEC 2** · Sec. 1.1 · p. 4
Came in	Sheet 1 · 1.1 · Midterm OLD · Model-C · Quiz 1 · Section

Chapter 1

★ Cheat sheet · Chapter 1

Remember

Q9 x(n) = A cos(ωn + θ) is periodic only for rational f, frequencies 2π apart are identical, and the fastest oscillation is at ω = π (f = 1/2).

Q15 t = nT gives F = f / Fs (and ω = Ω T): the digital frequency is the analog frequency divided by the sampling rate.

Q18 Each sample is mapped to one of L levels with step Δ = (xmax − xmin)/(L − 1); the error eq(n) is at most Δ/2 when rounding.

Q16 |F| ≤ 1/2 with F = f/Fs → f ≤ Fs/2 for every component → Fs ≥ 2 fmax; otherwise higher frequencies alias.

Q8 xa(t) = A cos(Ωt + θ) is always periodic (Tp = 1/F), distinct F give distinct signals, and a higher F means faster oscillation.

Q14 Periodic sampling takes x(n) = xa(nT), one sample every T = 1/Fs seconds.

Q22 Smaller Δ → more accurate (× as written); Nyquist rate = minimum sampling rate (✓); reconstruction needs Fs > $2f_o$ (× as written).

Q1 Signal = physical quantity varying with an independent variable; system = device that operates on a signal; processing = passing a signal through a system (e.g. filtering).

Equations

$$x(n) = x_a(nT) = A\cos(2\pi f n T + \theta) = A\cos(2\pi n f/F_s + \theta) \quad (1.32)$$

$$\mathbf{F = f / F_s} \quad (1.33) \qquad \text{or} \qquad \mathbf{\Omega = \omega T} \quad (1.34)$$

$$-1/2 \le f/F_s \le 1/2 \;\rightarrow\; -F_s/2 \le f \le F_s/2 \quad (1.37)$$

$$f_{max} = F_s/2 \;\rightarrow\; \mathbf{F_s \ge 2f_{max}}$$

SECTION 02

Chapter 1

Came once in the exams

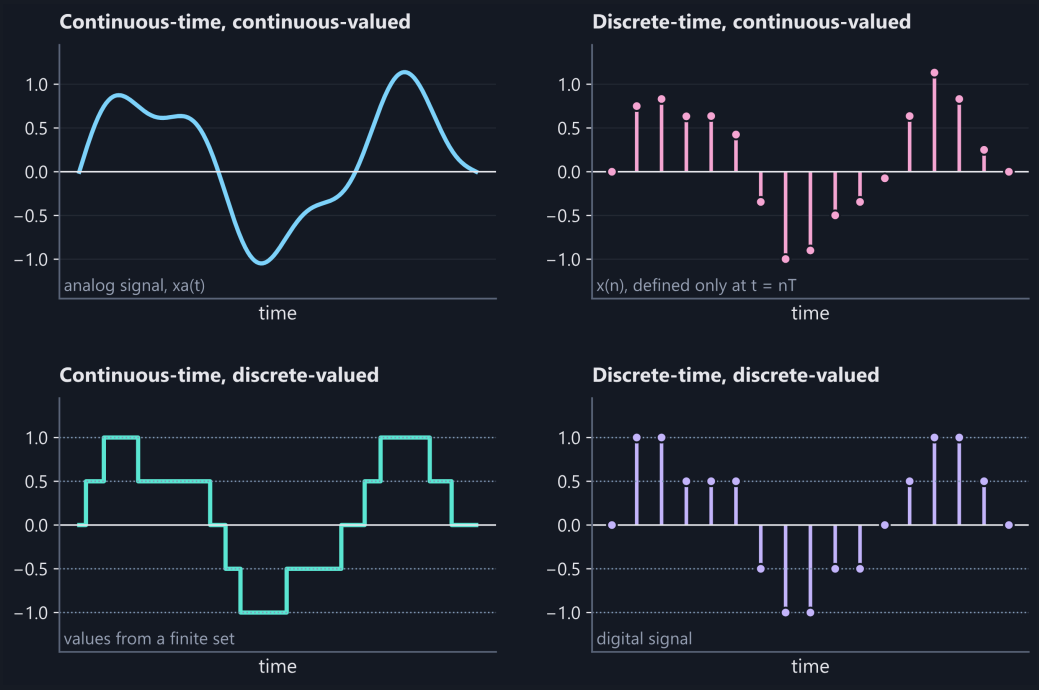
Questions — most frequent first

- ☐ Q4 Explain, with drawings, the different classifications of signals. p. 13
- ☐ Q13 Explain (with a block diagram) the basic three operations performed in the A/D conve... p. 14
- ☐ Q10 Show that a discrete-time sinusoid is periodic only if its frequency F is a rational... p. 16
- ☐ Q3 What are the advantages of digital signal processing over analog signal processing? ... p. 17
- ☐ Q5 Compare between deterministic and random signals. p. 18

Q4 ● Frequent 4×

Explain, with drawings, the different classifications of signals.

1. **Continuous-time vs. discrete-time:** continuous-time (analog) signals are defined for every value of time and take on values in the continuous interval (a, b). Discrete-time signals are defined only at certain specific values of time.
2. **Continuous-valued vs. discrete-valued:** a continuous-valued signal takes on all possible values on a finite or an infinite range. A discrete-valued signal takes on values from a finite set of possible values.
3. **Deterministic vs. random:** a deterministic signal can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. Otherwise, the signal is random.
4. **Multichannel vs. multidimensional:** a multichannel signal is generated by multiple sources or sensors and represented in vector form, $S3(t) = [s1(t)\ s2(t)\ s3(t)]^T$. A signal is M-dimensional if its value is a function of M independent variables.



Key idea احفظ التصنيفات كأزواج: continuous / discrete time، و continuous / discrete، valued، و deterministic / random، و multichannel / multidimensional.

Key words: Deterministic vs. random

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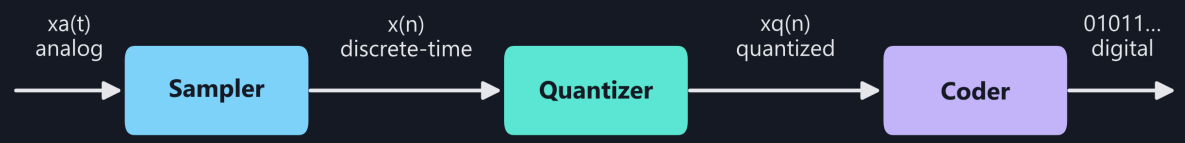
Lecture **LEC 2** · Sec. 1.4 · p. 18

Came in Midterm 8-12-2020 · Q2-a | Revision Sheet 1 | Sheet 1 · 1.3 | Midterm theory notes

Q13 · Frequent · 4×

Explain (with a block diagram) the basic three operations performed in the A/D converter.

To process analog signals by digital means, it is first necessary to convert them into a sequence of numbers having finite precision. This is A/D conversion, a three-step process (Fig. 1.9):



1. **Sampling:** the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. If xa(t) is the input to the sampler, the output is **x(n) = xa(nT)**, where T is the sampling interval.
2. **Quantization:** the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each sample is selected from a finite set of possible values. The difference $x_q(n) - x(n)$ is the quantization error.
3. **Coding:** each discrete value $x_q(n)$ is represented by a **b-bit binary sequence**.

Sampling does not result in a loss of information if the signal bandwidth is finite (**invertible**), while quantization is a **noninvertible** process that results in signal distortion.

Key idea الـ A/D تلات خطوات ورا بعض: الـ sampler بيقطّع الوقت، والـ quantizer بيقطّع القيم، والـ coder بيحوّل كل level لـ binary.

Key words Sampling Quantization Coding

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.6 · p. 3

Came in Revision Sheet 1 · Midterm OLD · Model-A 2012/2013 · Sheet 1 · 1.4 · Midterm theory notes

Q10 2×

Show that a discrete-time sinusoid is periodic only if its frequency F is a rational number. How do you find its fundamental period?

By definition, x(n) is periodic with period N (N > 0) if and only if x(n + N) = x(n) for all n (1.15). For a sinusoid with frequency F_o to be periodic:

$$\cos[2\pi F_0(N + n) + \theta] = \cos(2\pi F_0 n + \theta)$$

This relation is true if and only if there exists an integer k such that $2\pi F_o N = 2k\pi$, or equivalently:

$$F_0 = k / N \quad (1.16)$$

So F_o must be rational. To find the fundamental period, cancel common factors so that k and N are relatively prime; then the fundamental period is N.

Key idea عشان x(n + N) = x(n) لازم 2πFN = 2πk، يعني F = k/N لازم تبقى rational، وN بعد ما نختصر الكسر هي الـ fundamental period.

Source — from your PDFs

Lecture	**LEC 3** · Sec. 1.5.2 · p. 9
Came in	Midterm 1 Model-2 · Q2-b · Notebook / Section

Q3 1×

What are the advantages of digital signal processing over analog signal processing? What is its main limitation?

1. **Flexibility:** a digital programmable system allows reconfiguring the operations simply by changing the program; an analog system needs hardware redesign.
2. **Accuracy:** a digital system provides much better control of accuracy requirements; tolerances in analog components make accuracy hard to control.
3. **Storage:** digital signals are easily stored on magnetic media without deterioration, so they are transportable and can be processed off-line.
4. Allows the implementation of more **sophisticated** signal processing algorithms.
5. **Precise mathematical operations** are very difficult in analog form but routine on a digital computer using software.
6. In some cases a digital implementation is **cheaper** than its analog counterpart.

Limitation: the speed of operation of A/D converters and digital signal processors; signals with extremely wide bandwidths may be beyond the state of the art of digital hardware.

Key idea الـ DSP أحسن في الـ flexibility والـ accuracy والتخزين والـ algorithms المعقدة والتكلفة. عيبه سرعة الـ A/D والـ processor، ودي اللي بتحدد الـ bandwidth.

Key words Flexibility Accuracy Storage

Source — from your PDFs

Lecture	**LEC 2** · Sec. 1.3 · p. 13
Came in	Midterm OLD

Q5 1×

Compare between deterministic and random signals.

- **Deterministic:** any signal that can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. All past, present, and future values are known precisely, without any uncertainty.
- **Random:** otherwise, we refer to these signals as random.

Key idea الـ deterministic نقدر نوصفها بالظبط بمعادلة أو جدول ونعرف قيمتها في أي وقت. الـ random ما نقدرش نتوقع قيمتها بالظبط.

Key words Deterministic Random

Source — from your PDFs

Lecture	**LEC 2** · Sec. 1.4.3 · p. 26
Came in	Midterm OLD · Model-A 2013/2014

★ Cheat sheet · Chapter 1

Remember

Q3 DSP wins in flexibility, accuracy, storage, sophisticated algorithms and cost; its limitation is the speed of the A/D converters and processors (bandwidth).

Q4 Continuous vs discrete time, continuous vs discrete valued, deterministic vs random, multichannel vs multidimensional.

Q5 Deterministic = described exactly by an expression, table or rule (past, present and future known); random = cannot be predicted exactly.

Q10 $x(n + N) = x(n)$ needs $2\pi fN = 2\pi k$, so $f = k/N$ must be rational; N (with k/N in lowest terms) is the fundamental period.

Q13 A/D = sampler (discrete time) → quantizer (discrete values) → coder (b-bit binary words).

Equations

$$F_0 = k / N \quad (1.16)$$

SECTION 03

Chapter 2

Repeated in exams (2 times or more), most repeated first

Questions — most frequent first

- ☐ **Q3** Prove that the DFT, X(k), of a sequence x(n) is periodic. p. 19
- ☐ **Q8** Check (✓) or (×) and correct the mistakes (DFT / FT statements). p. 20

Q3 ● Frequent 13×

Prove that the DFT, X(k), of a sequence x(n) is periodic.

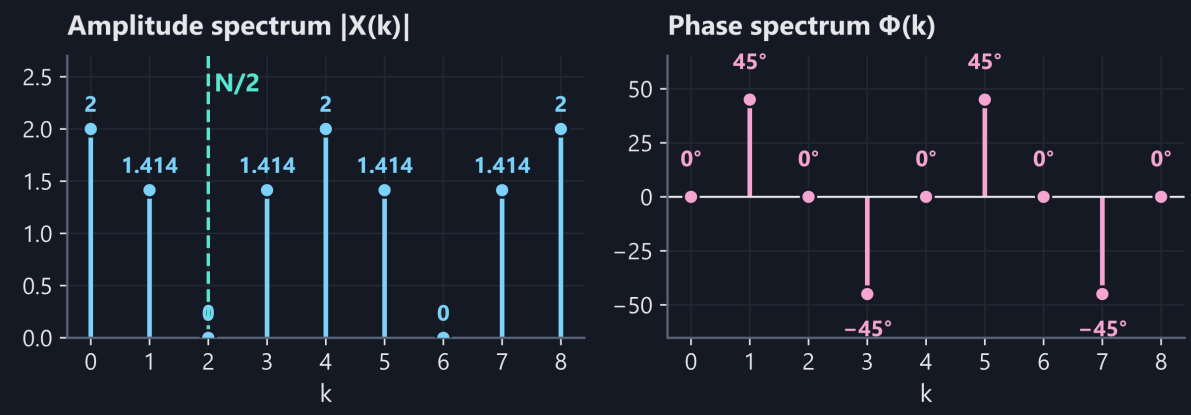
$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}$$

$$X(k+N) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N} \cdot e^{-j2\pi Nn/N} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N} \cdot e^{-j2\pi n}$$

Since n is integral, $e^{-j2\pi n} = 1$, so:

$$\mathbf{X(k+N) = X(k)}$$

The DFT is periodic with period N (the cyclical property of the DFT). E.g. for {1, 0, 0, 1}: X(0) = 2, therefore X(4) = 2 also.



Key idea لأن e^(−j2πn) = 1 لأي n صحيح، لما نعوّض k + N بنرجع لنفس القيمة، يعني X(k + N) = X(k) والـ DFT بيتكرر كل N.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.1 · p. 4

Came in Final 7-6-2023 · Q1-b | Final 9-6-2026 · Q2-a | Final 17-8-2026 · Q2-a | Final 23-1-2023 · Q3-a | Final 12-1-2019 · Q2-b | Final 2-1-2017 · Q2-b | Final 23-12-2017 · Q2-b | Midterm 12-11-2025 · Q2-a | Midterm 28-11-2021 · Q2-a | Midterm 8-12-2020 · Q2-b | Midterm 1 Model-2 · Q2-a | Midterm 2 · Q2-b | Sheet 2 · 2.1

Q8 3×

Check (✓) or (×) and correct the mistakes (DFT / FT statements).

1. Fourier transform (FT) is calculated for non-periodic continuous-time signals. **(✓)**
2. Discrete Fourier transform (DFT) is obtained for the analog signals. **(×)**
   Correct: the DFT is obtained for the sampled (discrete-time) sequence x(nT).

Key idea الـ FT للإشارات الـ continuous اللي مش periodic، فدي صح. والـ DFT بيتحسب على الـ samples مش على الإشارة الأنالوج، فالتانية غلط.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.1 · p. 4

Came in Final 12-1-2019 · Q5 | Final 23-12-2017 · Q5-b | Quiz 1 Summer 26 · Q3

★ Cheat sheet · Chapter 2

Remember

Q3 DSP wins in flexibility, accuracy, storage, sophisticated algorithms and cost; its limitation is the speed of the A/D converters and processors (bandwidth).

Q8 $x_a(t) = A\cos(\Omega t + \theta)$ is always periodic ($T_p = 1/F$), distinct F give distinct signals, and a higher F means faster oscillation.

SECTION 04

Chapter 2

Came once in the exams

Questions — most frequent first

☐ **Q6** State the mathematical properties of the DFT. p. 22

Q6 1×

State the mathematical properties of the DFT.

1. **Symmetry:** Re[X(N − k)] = Re X(k) (2.8) and Im[X(N − k)] = −Im[X(k)] (2.9).
2. **Even functions:** if $x_e(n) = x_e(-n)$: $X_e(k) = \Sigma\, x_e(n) \cos(k\Omega nT)$ (2.10).
3. **Odd functions:** if $x_o(n) = -x_o(-n)$: $X_o(k) = -j\, \Sigma\, x_o(n) \sin(k\Omega nT)$ (2.11).
4. **Parseval's theorem:** $\Sigma\, x^2(n) = (1/N)\, \Sigma\, |X(k)|^2$ (2.12).
5. **Delta function:** $F_D[\delta(nT)] = 1$ (2.13).
6. Linear cross-correlation of two sequences may be computed using DFTs.
7. DFTs may be used to compute circular convolutions and, using augmenting zeros, linear convolutions.

Key idea احفظهم كقايمة: symmetry، even و odd functions، Parseval و delta، function، linear cross-correlation و circular convolution.

Key words: Symmetry · Even functions · Odd functions · Parseval's theorem

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Lecture	**LEC 5** · Sec. 2.3 · p. 27
Came in	Final 18-1-2024 · Q2-a

★ Cheat sheet · Chapter 2

Remember

Q6 A digital signal is discrete in time and in amplitude; a digital processor can only handle sampled and quantized values.